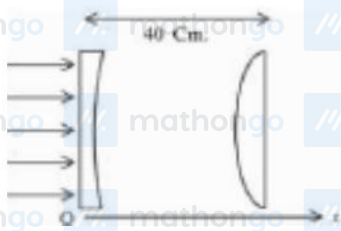


Q1 - 24 January - Shift 1

As shown in the figure, a combination of a thin plano concave lens and a thin plano convex lens is used to image an object placed at infinity. The radius of curvature of both the lenses is 30 cm and refractive index of the material for both the lenses is 1.75. Both the lenses are placed at distance of 40 cm from each other. Due to the combination, the image of the object is formed at distance $x =$ _____ cm, from concave lens.

*Space for your notes:***Q2 - 24 January - Shift 2**

When a beam of white light is allowed to pass through convex lens parallel to principal axis, the different colours of light converge at different point on the principle axis after refraction. This is called :

- (1) Scattering
- (2) Chromatic aberration
- (3) Spherical aberration
- (4) Polarisation

*Space for your notes:***Q3 - 24 January - Shift 2**

A convex lens of refractive index 1.5 and focal length 18 cm in air is immersed in water. The change in focal length of the lens will be _____ cm.

(Given refractive index of water = $\frac{4}{3}$)

Space for your notes:

Q4 - 25 January - Shift 1

A ray of light is incident from air on a glass plate having thickness $\sqrt{3}$ cm and refractive index $\sqrt{2}$. The angle of incidence of a ray is equal to the critical angle for glass-air interface. The lateral displacement of the ray when it passes through the plate is $___\times 10^{-2}$ cm. (given $\sin 15^\circ = 0.26$)

Space for your notes:

Q5 - 25 January - Shift 2

The light rays from an object have been reflected towards an observer from a standard flat mirror, the image observed by the observer are :-

- A. Real
- B. Erect
- C. Smaller in size than object
- D. Laterally inverted

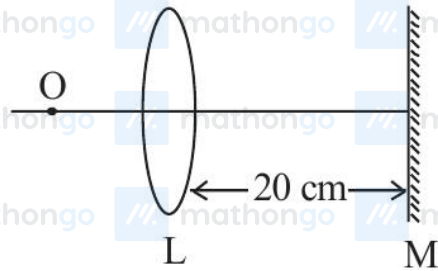
Choose the most appropriate answer from the options given below :

- (1) B and D only
- (2) B and C only
- (3) A and D only
- (4) A, C and D only

Space for your notes:

Q6 - 25 January - Shift 2

An object is placed on the principal axis of convex lens of focal length 10 cm as shown. A plane mirror is placed on the other side of lens at a distance of 20 cm. The image produced by the plane mirror is 5 cm inside the mirror. The distance of the object from the lens is _____ cm.



Space for your notes:

Q7 - 29 January - Shift 2

A scientist is observing a bacteria through a compound microscope. For better analysis and to improve its resolving power he should. (Select the best option)

- (1) Increase the wave length of the light
- (2) Increase the refractive index of the medium between the object and objective lens
- (3) Decrease the focal length of the eye piece
- (4) Decrease the diameter of the objective lens

Space for your notes:

Q8 - 30 January - Shift 1

A person has been using spectacles of power-1.0 diopter for distant vision and a separate reading glass of power 2.0 diopters. What is the least distance of distinct vision for this person:

- (1) 10 cm
- (2) 40 cm
- (3) 30 cm
- (4) 50 cm

Space for your notes:

Q9 - 30 January - Shift 1

In an experiment for estimating the value of focal length of converging mirror, image of an object placed at 40 cm from the pole of the mirror is formed at distance 120 cm from the pole of the mirror. These distances are measured with a modified scale in which there are 20 small divisions in 1 cm. The value of error in measurement of focal length of the mirror is $1/K$ cm. The value of K is _____.

Space for your notes:

Q10 - 30 January - Shift 2

A thin prism P_1 with an angle 6° and made of glass of refractive index 1.54 is combined with another prism P_2 made from glass of refractive index 1.72 to produce dispersion without average deviation.

The angle of prism P_2 is :

- (1) 6° (2) 1.3°
(3) 7.8° (4) 4.5°

Space for your notes:

Q11 - 31 January - Shift 1

In a medium the speed of light wave decreases to 0.2 times to its speed in free space. The ratio of relative permittivity to the refractive index of the medium is $x : 1$. The value of x is _____.

(Given speed of light in free space $= 3 \times 10^8 \text{ m s}^{-1}$ and for the given medium $\mu_r = 1$)

Space for your notes:

Q12 - 31 January - Shift 2

A microscope is focused on an object at the bottom of a bucket. If liquid with refractive index $\frac{5}{3}$ is poured inside the bucket, then microscope have to be raised by 30 cm to focus the object again. The height of the liquid in the bucket is :

- (1) 75 cm (2) 50 cm
(3) 18 cm (4) 12 cm

Space for your notes:

Q13 - 01 February - Shift 1

A thin cylindrical rod of length 10 cm is placed horizontally on the principle axis of a concave mirror of focal length 20 cm. The rod is placed in a such a way that mid point of the rod is at 40 cm from the pole of mirror. The length of the image formed by the mirror will be $\frac{x}{3}$ cm. The value of x is

Space for your notes:

Q14 - 01 February - Shift 2

Two objects A and B are placed at 15 cm and 25 cm from the pole in front of a concave mirror having radius of curvature 40 cm. The distance between images formed by the mirror is:

- (1) 40 cm (2) 60 cm
(3) 160 cm (4) 100 cm

Space for your notes:

Answer Key

Q13 (32) **Q14 (3)**

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Q1 (120)

$$\frac{1}{f_1} = (1.75 - 1) \left(-\frac{1}{30} \right)$$

$$\Rightarrow f_1 = -40 \text{ cm}$$

$$\frac{1}{f_2} = (1.75 - 1) \left(\frac{1}{30} \right) \Rightarrow f_2 = 40 \text{ cm}$$

Image from L_1 will be virtual and on the left of L_1 at focal length 40 cm. So the object for L_2 will be 80 cm from L_2 which is $2f$. Final image is formed at 80 cm from L_2 on the right.

So $x = 120$

Q2 (2)

Based on fact

Q3 (54)

$$\frac{I}{f_{\text{H}_2\text{O}}} = \left(\frac{\mu_g}{\mu_{\text{H}_2\text{O}}} - 1 \right) \left(\frac{2}{R} \right)$$

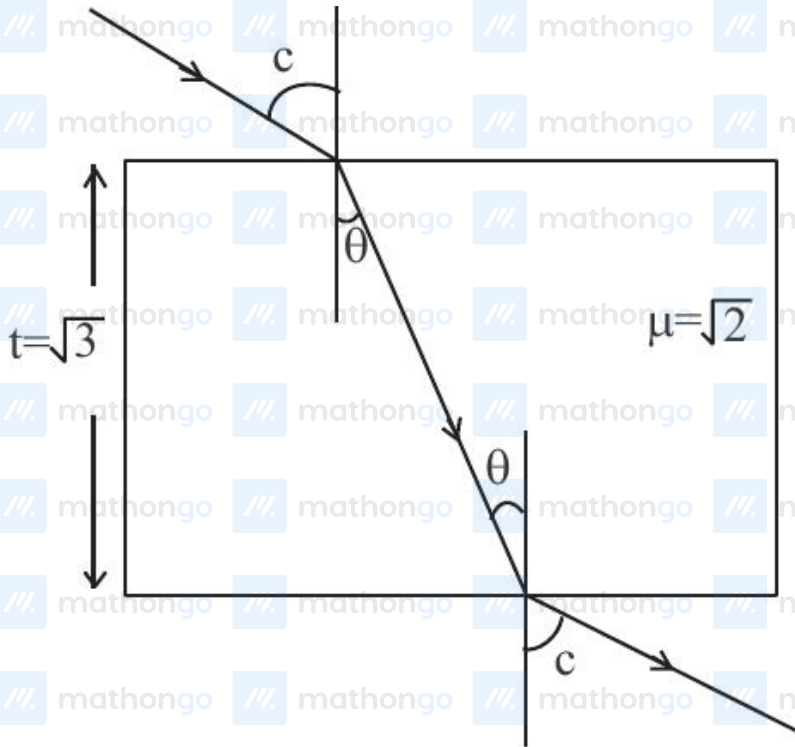
$$= \frac{1}{8} \left(\frac{2}{R} \right)$$

$$= \frac{1}{(4f_{\text{air}})}$$

$$\text{So, } f_{\text{H}_2\text{O}} = 4f_{\text{air}} = 72 \text{ cm}$$

$$\text{So change in focal length} = 72 - 18 = 54 \text{ cm}$$

Q4 (52)



$$\sin c = \frac{1}{\sqrt{2}}$$

$$c = 45^\circ$$

$$\sin c = \mu \sin \theta$$

$$\frac{1}{\sqrt{2}} = \sqrt{2} \sin \theta$$

$$\theta = 30^\circ$$

Lateral displacement:

$$x = t \sin(i - r) \sec r$$

$$x = \sqrt{3} \sin(45^\circ - 30^\circ) \sec 30^\circ$$

$$x = \sqrt{3} (0.26) \left(\frac{2}{\sqrt{3}} \right)$$

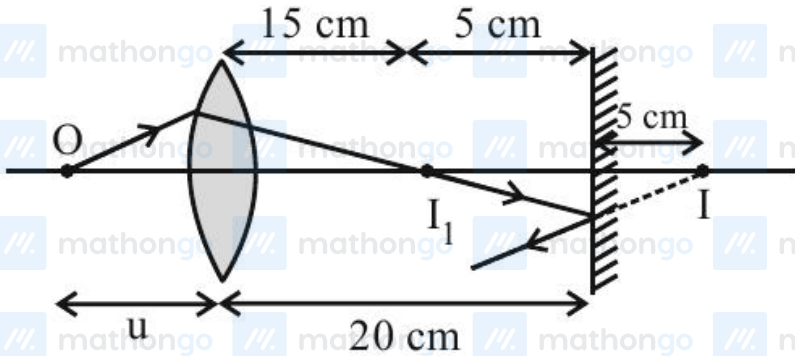
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$$X = 0.52 \text{ cm}$$

Q5 (1)

Plane mirror forms erect, same sized, laterally inverted and virtual image of real object.

Q6 (30)



$$f = 10 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{15} - \frac{1}{-u} = \frac{1}{10}$$

$$\Rightarrow \frac{1}{u} = \frac{1}{10} - \frac{1}{15}$$

On solving we get value of u as 30 cm.

Q7 (1)

$$P = \frac{2\mu \sin \theta}{1.22\lambda}$$

Q8 (4)

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f},$$

$$P = 2D = 2\text{m}^{-1}$$

$$\Rightarrow \frac{1}{f} = \frac{2}{100} \text{cm}^{-1}$$

$$\frac{1}{V} - \left(-\frac{1}{25} \right) = \frac{2}{100}$$

$$\Rightarrow \frac{1}{V} = \frac{1}{50} - \frac{1}{25}$$

$$\Rightarrow V = -50 \text{ cm}$$

Q9 (32)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$
$$\frac{-1}{120} - \frac{1}{40} = \frac{1}{f}, \quad f = -30 \text{ cm}$$

Now,

$$\frac{-1}{v^2} dv - \frac{1}{u^2} du = -\frac{1}{f^2} df$$

$$\text{Also } dv = du = \frac{1}{20} \text{ cm}$$

$$\frac{1}{(120)^2} + \frac{1}{(40)^2} = \frac{df}{(30)^2}$$

On solving

$$df = \frac{1}{32} \text{ cm}$$

$$\therefore k = 32$$

Q10 (4)

$$\delta_1 = \delta_2 \text{ [for no average deviation]}$$

$$\Rightarrow 6^\circ (1.54 - 1) = A(1.72 - 1)$$

$$\Rightarrow A = \frac{6^\circ \times 0.54}{0.72}$$

$$= \frac{18^\circ}{4} = 4.5^\circ$$

Q11 (5)

$$V = \frac{C}{\mu} \Rightarrow \mu = \frac{C}{V} = \frac{C}{0.2C}$$

$$\boxed{\mu = 5}$$

$$\mu = \sqrt{\epsilon_r \mu_r}$$

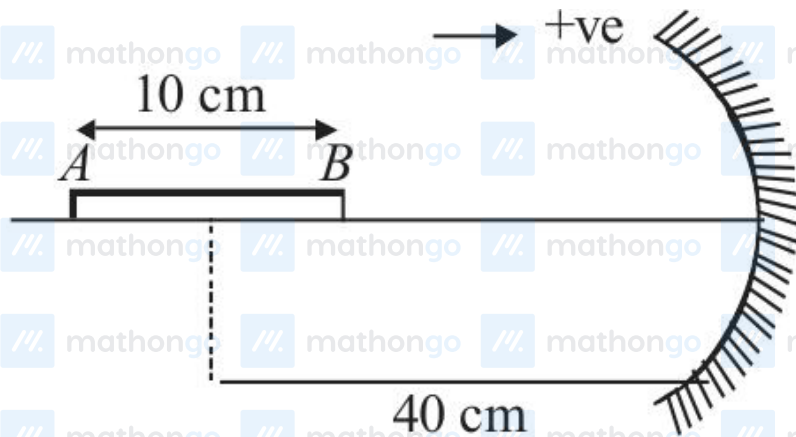
$$\Rightarrow \epsilon_r = \frac{\mu^2}{\mu_r}$$

$$\therefore \epsilon_r = \frac{\mu}{1} = 5$$

Q12 (1)

No Solution

Q13 (32)



$$U_A = -45 \text{ cm}, f = -20 \text{ cm}$$

$$V_A = \frac{-45 \times (-20)}{-45 - (-20)} = \frac{-900}{-25} = -36 \text{ cm}$$

$$\text{And } U_B = -35 \text{ cm}$$

$$\therefore V_B = \frac{-35 \times (-20)}{-35 - (-20)} = \frac{700}{-15}$$

$$\therefore V_A - V_B = \text{length of image}$$

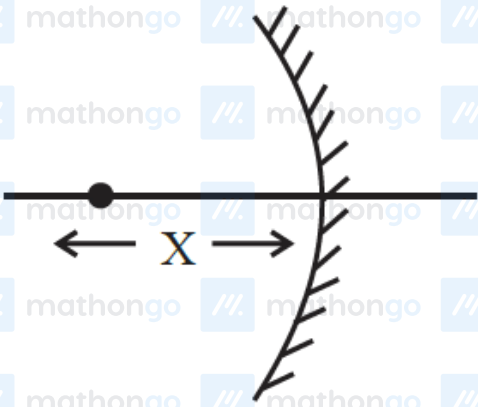
$$= \left(-36 + \frac{140}{3} \right) \text{ cm}$$

$$= \frac{-108 + 140}{3} \text{ cm}$$

$$= \frac{32}{3} \text{ cm}$$

$$\therefore x = 32$$

Q14 (3)



by mirror formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v_1} + \frac{1}{-15} = \frac{1}{(-20)}$$

$$\frac{1}{v_1} = -\frac{1}{20} + \frac{1}{15}$$

$$= \frac{-3 + 4}{60}$$

$$v_1 = 60 \text{ cm}$$

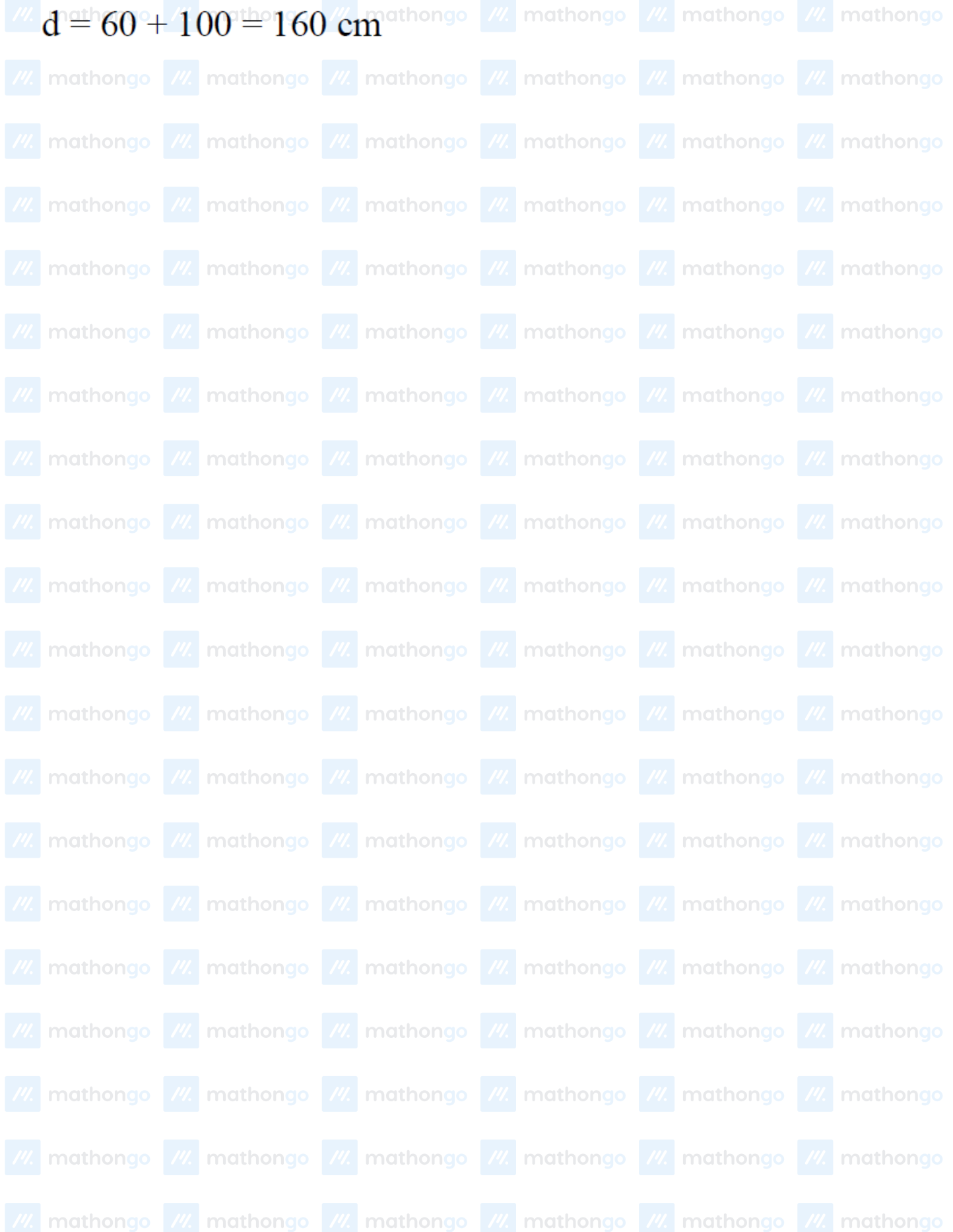
$$\frac{1}{v_2} + \frac{1}{(-25)} = \frac{1}{(-20)}$$

$$\frac{1}{v_2} = \frac{-1}{20} + \frac{1}{25}$$

$$= \frac{-5 + 4}{100} = \frac{-1}{100}$$

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$$v_2 = -100 \text{ cm}$$
$$d = 60 + 100 = 160 \text{ cm}$$



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